A Cosine-Projection Framework for Choosing Among Cox HR, D-RMST, and Average-Hazard Tests in Survival Trials

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Abstract

The log-rank/Cox hazard-ratio test remains the standard approach for the primary analysis of randomized trials with time-to-event endpoints and is fully efficient under proportional hazards (PH). Two model-free alternatives that have recently gained attention are the difference in restricted mean survival time (D-RMST), a contrast on the time scale, and the more recently proposed ratio of average hazards (RAH), a contrast on the rate scale. Simulation studies suggest that the RAH-based test can outperform both under delayed treatment effects, but its efficiency has not been characterized analytically, and no unified framework explains how censoring and accrual affect the efficiency ordering among these tests. We show that all three statistics share a weighted-log-rank representation and derive their asymptotic relative efficiencies under local alternatives, including the first analytic efficiency results for the RAH-based test. Geometrically, each test defines a direction in a function space and the anticipated treatment effect defines a signal direction. A test's efficiency relative to the oracle test equals the squared cosine of the angle between these directions, so the asymptotic relative efficiency of any two tests is a ratio of squared cosines. Because censoring and accrual enter through an information weight, the framework shows how design choices can alter, and even reverse, the efficiency ordering. We illustrate the framework using the CheckMate 057 immunotherapy trial and corroborate the theoretical results through simulations spanning proportional- and non-proportional-hazards settings.

1 Introduction

Non-proportional hazards arise in randomized clinical trials with time-to-event endpoints, particularly in immuno-oncology, where treatment effects may be delayed and hazards may cross [Horiguchi et al., 2019, Uno and Horiguchi, 2023]. In such settings, the choice of the primary analysis method is an important design decision. This paper compares three tests that a trialist might reasonably consider. The first is the conventional test based on the Cox proportional hazards model, asymptotically equivalent to the log-rank test. For the primary analysis of overall survival, the FDA recommends the log-rank test, accompanied by the hazard ratio from a Cox model as the summary measure [U.S. Food and Drug Administration,

2025], and the test is asymptotically fully efficient when the proportional hazards (PH) assumption holds. The second is the test based on the difference in restricted mean survival time (D-RMST) up to a prespecified time horizon, an established alternative that compares survival on the time scale without requiring the PH assumption [Royston and Parmar, 2013, Uno et al., 2014, Tian et al., 2014, Uno et al., 2022]. The third is the test based on the ratio of average hazards (RAH), proposed recently by Uno and Horiguchi [2023]. Average hazard summarizes the population-level event rate per unit of event-free person-time over a prespecified time horizon, and the RAH compares these rates between groups. Like D-RMST, RAH is defined without requiring the PH assumption, but it summarizes the between-group contrast on the rate scale rather than the time scale [Uno and Horiguchi, 2023, Meng et al., 2026].

The choice among these tests can have substantial consequences at the design stage. Power calculations for time-to-event endpoints require specification of the baseline survival distribution, the anticipated treatment-effect pattern, the censoring and accrual distributions, and the analysis time horizon. The relative efficiency of the Cox-, D-RMST-, and RAH-based tests therefore depends on these design features. This motivates the practical question: *given these design specifications, which of the three tests is the most efficient?*

Previous work has examined this question from both theoretical and empirical perspectives. On the theoretical side, Tian et al. [2018] compared the log-rank and D-RMST tests through their asymptotic relative efficiency (ARE) under local alternatives. They showed that the D-RMST test is nearly as efficient as the log-rank test under PH, can be more efficient under some non-PH alternatives, but can lose efficiency when the treatment effect emerges primarily later in follow-up. On the empirical side, Uno and Horiguchi [2023] compared average-hazard-based tests with the Cox- and RMST-based tests through simulation under PH, early-difference, and delayed-difference patterns and under different censoring distributions. Their results showed that the relative performance of the tests can vary substantially with both the treatment-effect pattern and the censoring distribution. Despite these developments, the RAH-based test has not been placed on the same theoretical efficiency scale as the Cox- and D-RMST-based tests. Its power properties have been documented by simulation, but no analytic expression for its efficiency is available, and no unified framework explains when and why the efficiency ordering among the three tests arises or how censoring and accrual can alter that ordering.

We develop such a unified theoretical framework. We first show that all three statistics admit a common weighted-log-rank representation and derive their asymptotic relative efficiencies under local alternatives, including the first analytic efficiency results for the RAH-based test. Although the resulting weight functions indicate where each test places emphasis over follow-up, their shapes alone do not determine relative efficiency because the amount of information available at each time also depends on the trial design. We therefore recast the comparison geometrically. Each test corresponds to a direction in a function space determined jointly by its weight function and the design's information weight, while the treatment-effect pattern defines a signal direction in the same space. The efficiency of a test relative to the oracle test for that alternative is exactly the squared cosine of the angle between these two directions, and the ARE of any two tests is therefore a ratio of squared cosines. This *cosine metric* turns efficiency comparisons into statements about alignment and makes explicit how censoring and accrual can alter, and even reverse, the efficiency

ordering among tests.

We evaluate the theoretical results through simulations under PH and non-PH scenarios, including fading and delayed treatment effects, and illustrate the design-stage use of the framework in a case study based on reconstructed CheckMate 057 data. These analyses assess how well the theoretical efficiency comparisons predict empirical performance across different treatment-effect patterns and trial designs.

The remainder of the paper is organized as follows. Section 2 defines the three test statistics, derives their common weighted-log-rank representation, and presents their asymptotic efficiencies under local alternatives. Section 3 develops the cosine metric and its geometric interpretation. Section 4 reports the simulation studies, and Section 5 presents the CheckMate 057 design analysis. Section 6 concludes with practical implications and directions for future work.

2 Three Test Statistics, a Weighted-Log-Rank Representation, and Their Efficiency

2.1 Setup and notation

Let T_k $(k = 0, 1)$ denote the event time in group k with survival function $S_k(t)$, cumulative hazard function $\Lambda_k(t)$, and hazard function $\lambda_k(t)$. Define

$$F_k(t) = 1 - S_k(t) \qquad \text{and} \qquad R_k(t) = \int_0^t S_k(u)\, du. \tag{1}$$

Thus, $F_k(t) = \Pr(T_k \le t)$ is the cumulative incidence rate (the probability that the event has occurred by time t) and $R_k(\tau)$ is the RMST on $(0, \tau]$. Let C_k be the censoring time, independent of T_k, with $G_k(t) = \Pr(T_k \wedge C_k \ge t)$ bounded away from zero on $(0, \tau]$. Let p_k be the limiting proportion of subjects in group k $(p_0 + p_1 = 1)$. Lastly, the at-risk weight function is defined as

$$w(t) = \frac{p_0\, p_1\, G_0(t)\, G_1(t)}{p_0\, G_0(t) + p_1\, G_1(t)}. \tag{2}$$

Two simplifying conditions recur throughout the paper. By *equal allocation* we mean a balanced design with $p_0 = p_1 = 1/2$. By *a common censoring distribution* we mean that the two groups share the same marginal censoring law, and we write $S_C(v) = \Pr(C \ge v)$ for the common censoring survival function; this is the case for administrative censoring. Under these two conditions, the at-risk weight (2) becomes $w(t) = 0.5 \times S_C(t)/\{S_0(t)^{-1} + S_1(t)^{-1}\}$. If the null hypothesis, $S_1(t) = S_0(t), t \in [0, \tau]$, is also true, then the weight function further reduces to $w(t) = S_0(t)S_C(t)/4$.

2.2 The three test statistics

We can test $H_0 : S_1(t) = S_0(t), t \in [0, \tau]$ against alternatives using three statistics. For each statistic, we state the corresponding estimand, point estimator, and asymptotic distribution

of the test statistic. Throughout, $\hat{S}_k(\cdot)$ and $\hat{\Lambda}_k(\cdot)$ denote the Kaplan–Meier and Nelson–Aalen estimators of $S_k(\cdot)$ and $\Lambda_k(\cdot)$, respectively, $\hat{G}_k(\cdot)$ denote the empirical counterpart of $G_k(\cdot)$, and $\hat{R}_k(t)$ denote $\int_0^t \hat{S}_k(u)du$, a consistent estimator of $R_k(t)$.

Cox's HR-based test. Under the PH assumption,

$$\theta = \log\left\{\frac{\lambda_1(t)}{\lambda_0(t)}\right\}$$

does not depend on t and can be consistently estimated by the maximum partial likelihood estimator $\hat{\theta}$. As $n \to \infty$,

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \sigma_\theta^2),$$

where

$$\sigma_\theta^2 = \left\{\int_0^\tau \frac{e^\theta p_0 p_1 \, G_0(u) \, G_1(u)}{p_0 \, G_0(u) + e^\theta p_1 \, G_1(u)} \, d\Lambda_0(u)\right\}^{-1}. \tag{3}$$

Under $H_0 : \lambda_1(\cdot) = \lambda_0(\cdot)$,

$$\sigma_\theta^2 = \left\{\int_0^\tau w(u) \, d\Lambda_0(u)\right\}^{-1},$$

and the Wald statistic $Z_{\text{Cox}} = \sqrt{n}\,\hat{\theta}/\hat{\sigma}_\theta$ is asymptotically standard normal, where $\hat{\sigma}_\theta^2$ is a consistent estimator of σ_θ^2.

We use Z_{Cox} for all numerical studies in this paper, whereas its theoretical properties were examined based on an asymptotically equivalent score statistic, i.e., the log-rank statistic. The details are given in Section 2.3 and Appendix B.

D-RMST. The **D-RMST** test is based on the difference between two estimated RMSTs:

$$\hat{\theta}_{D\text{-RMST}} = \hat{R}_1(\tau) - \hat{R}_0(\tau) = \int_0^\tau \left[\hat{S}_1(t) - \hat{S}_0(t)\right] dt, \tag{4}$$

the area between the two Kaplan–Meier curves up to τ [Uno et al., 2014, 2022]. Let $\theta_{D\text{-RMST}} = R_1(\tau) - R_0(\tau)$ be the true RMST difference. It is a clinically interpretable estimand, and oftentimes preferred when the PH assumption may be violated. Then, $\hat{\theta}_{D\text{-RMST}}$ is a consistent estimator of $\theta_{D\text{-RMST}}$ and $\sqrt{n}(\hat{\theta}_{D\text{-RMST}} - \theta_{D\text{-RMST}}) \xrightarrow{d} N(0, \sigma^2_{D\text{-RMST}})$ with

$$\sigma^2_{D\text{-RMST}} = \sum_{k=0}^{1} \int_0^\tau \left\{\int_v^\tau S_k(u) \, du\right\}^2 \frac{d\Lambda_k(v)}{p_k \, G_k(v)},$$

as $n \to \infty$ [Tian et al., 2018]. Under the null, the Wald statistic $Z_{D\text{-RMST}} = \sqrt{n}\,\hat{\theta}_{D\text{-RMST}}/\hat{\sigma}_{D\text{-RMST}}$ is asymptotically standard normal, where $\hat{\sigma}^2_{D\text{-RMST}}$ is a consistent estimator of $\sigma^2_{D\text{-RMST}}$.

RAH. The AH for group k on $(0, \tau]$ is

$$\eta_k(\tau) = \frac{F_k(\tau)}{R_k(\tau)} = \int_0^\tau \lambda_k(t) \, \frac{S_k(t)}{\int_0^\tau S_k(u)\, du} \, dt,$$

a weighted average of hazard function over $[0, \tau]$ [Uno and Horiguchi, 2023]. $\eta_k(\tau)$ is a population-level person-time event rate. Two properties make AH attractive as a robust summary on the hazard scale. First, $\eta_k(\tau)$ is defined purely through the event-time distribution and is therefore not affected by the censoring distribution. On the other hand, the commonly used person-time incidence rate $\Pr(T \le C)/\mathbb{E}(T \wedge C)$ depends on both survival and censoring distributions [Uno and Horiguchi, 2023]. Second, AH respects stochastic ordering: if $S_1(t) \ge S_0(t)$ for all $t \le \tau$, then $\eta_1(\tau) \le \eta_0(\tau)$. The ratio of AH (RAH) between two groups is

$$\theta_{\mathrm{RAH}} = \frac{\eta_1(\tau)}{\eta_0(\tau)} = \frac{F_1(\tau)/R_1(\tau)}{F_0(\tau)/R_0(\tau)}, \tag{5}$$

with $\theta_{\mathrm{RAH}} = 1$ under H_0. Its estimator $\hat{\theta}_{\mathrm{RAH}}$ can be obtained by replacing $F_k(\tau)$ and $R_k(\tau)$ with $1 - \hat{S}_k(\tau)$ and $\hat{R}_k(\tau)$ in (5), respectively. As $n \to \infty$, $\sqrt{n}(\log \hat{\theta}_{\mathrm{RAH}} - \log \theta_{\mathrm{RAH}}) \xrightarrow{d} N(0, \sigma^2_{\mathrm{RAH}})$, [Uno and Horiguchi, 2023], where

$$\sigma^2_{\mathrm{RAH}} = \sum_{k=0}^{1} \int_0^\tau \left\{ \frac{1}{F_k(\tau)} - \frac{R_k(u)}{R_k(\tau)} \right\}^2 \frac{d\Lambda_k(u)}{G_k(u)\, p_k}. \tag{6}$$

The test statistic $Z_{\mathrm{RAH}} = \sqrt{n} \log(\hat{\theta}_{\mathrm{RAH}})/\hat{\sigma}_{\mathrm{RAH}}$ is asymptotically $N(0,1)$ under H_0, where $\hat{\sigma}^2_{\mathrm{RAH}}$ is a consistent estimator of σ^2_{RAH}.

2.3 Weighted-log-rank representation

All three statistics, to the first order, can be written as a weighted contrast of Nelson–Aalen cumulative-hazard estimates from two groups:

$$U_W = \int_0^\tau W(t) d\hat{\Delta}_\Lambda(t), \tag{7}$$

where

$$\hat{\Delta}_\Lambda(v) = \hat{\Lambda}_1(v) - \hat{\Lambda}_0(v).$$

Under H_0, the two cumulative hazards coincide and U_W has mean zero. The weight $W(\cdot)$ decides which time periods contribute to the comparison, and choosing $W(\cdot)$ to match the anticipated alternative is the classical route to constructing powerful, distribution-free survival tests [Fleming and Harrington, 1991, Tian et al., 2018]. Rigorous arguments for expressing three aforementioned test statistics in this form can be found in Appendix B.

Cox. The log-rank statistic, the score statistic under Cox model, can be written as

$$U_{\mathrm{Cox}} = \int_0^\tau \frac{p_1 p_0 \hat{G}_1(t) \hat{G}_0(t)}{p_1 \hat{G}_1(t) + p_0 \hat{G}_0(t)} d\hat{\Delta}_\Lambda(t). \tag{8}$$

Under the local alternatives considered in section 2.4, a one-step expansion of the score function from the partial likelihood function around $\theta = 0$ yields that $\sqrt{n}\,\hat{\theta} = \sigma_\theta^2 \sqrt{n}\, U_{\text{Cox}} + o_p(1)$. Thus, the Wald statistic, Z_{Cox} and the normalized $\sqrt{n}U_{\text{Cox}}$ are asymptotically equivalent. Furthermore, the random weight in (8) converges to at-risk weight $w(\cdot)$ in (2) in probability, as $n \to \infty$,

D-RMST and RAH. Neither statistic is a log-rank type statistic by construction, but each is a smooth functional of $\hat{\Lambda}_1(\cdot)$ and $\hat{\Lambda}_0(\cdot)$. For D-RMST, $R(\Lambda) = \int_0^\tau \exp\{-\Lambda(t)\}\, dt$, and

$$\begin{aligned}
\hat{\theta}_{D\text{-RMST}} &= \int_0^\tau S_0(t) \left[\exp\left\{-\hat{\Delta}_\Lambda(t)\right\} - 1\right] dt + O\big(\|\hat{\Delta}_\Lambda\|^2\big) \\
&= -\int_0^\tau S_0(t)\, \hat{\Delta}_\Lambda(t)\, dt + O\big(\|\hat{\Delta}_\Lambda\|^2\big) \\
&= -\int_0^\tau \{R_0(\tau) - R_0(t)\}\, d\hat{\Delta}_\Lambda(t) + O\big(\|\hat{\Delta}_\Lambda\|^2\big) \\
&= -\, U_{D\text{-RMST}} + O_p\big(n^{-1}\big)
\end{aligned}$$

under local alternative. As the remainder is $O_p(n^{-1})$, $Z_{D\text{-RMST}}$ and the normalized $\sqrt{n}U_{D\text{-RMST}}$ are asymptotically equivalent. RAH can be handled the same way, with an extra delta-method step for the ratio $F_k(\tau)/R_k(\tau)$, and

$$\begin{aligned}
\log(\hat{\theta}_{\text{RAH}}) =& \int_0^\tau \left[\frac{S_0(\tau)}{F_0(\tau)} + \frac{R_0(\tau) - R_0(t)}{R_0(\tau)}\right] d\hat{\Delta}_\Lambda(t) + O\big(\|\hat{\Delta}_\Lambda\|^2\big) \\
=& U_{\text{RAH}} + O_p(n^{-1})
\end{aligned}$$

under local alternative. As a result, Z_{RAH} and the normalized $\sqrt{n}U_{\text{RAH}}$ are asymptotically equivalent.

All U_X are oriented so that $U_X < 0$ indicates benefit of treatment (hence $U_{D\text{-RMST}} = -\hat{\theta}_{D\text{-RMST}}$ to first order); since ξ_X depends on μ_X^2, the orientation does not affect efficiency.

In summary, under equal allocation and a common censoring assumption, the resulting weights for U_{Cox}, $U_{D\text{-RMST}}$ and U_{RAH} are

$$\begin{aligned}
W_{\text{Cox}}(t) =&\ S_0(t)\, S_C(t)/4, \\
W_{D\text{-RMST}}(t) =& R_0(\tau) - R_0(t), \text{ and} \\
W_{\text{RAH}}(t) =& \left[\frac{S_0(\tau)}{F_0(\tau)} + \frac{R_0(\tau) - R_0(t)}{R_0(\tau)}\right],
\end{aligned}$$

respectively. Since multiplying the weight function by any nonzero constant yields an equivalent test statistic, without loss of generality, we may use following equivalent weight functions

$$\begin{aligned}
W_{\text{Cox}}(t) =& S_0(t)\, S_C(t) \\
W_{D\text{-RMST}}(t) =& \frac{R_0(\tau) - R_0(t)}{R_0(\tau)} \\
\text{and}\ \ W_{\text{RAH}}(t) =& S_0(\tau) + \{1 - S_0(\tau)\}\, W_{D\text{-RMST}}(t)
\end{aligned}$$

for three test statistics. On the current scale, all three weights equal 1 at $t = 0$, their limits at $t = \tau$ are $S_0(\tau)S_C(\tau)$, 0, and $S_0(\tau)$. Also, weight for U_{RAH} is a simple linear transformation of that for $U_{D\text{-RMST}}$.

The Cox model-based log-rank weight tracks the amount of information in the risk set. It is the largest early in follow-up, when the risk set is large, and decreases over time as deaths and censoring deplete the risk set. Under the PH assumption, this weighting is optimal. The D-RMST weight represents the RMST within the window $[t, \tau]$. It also places greater weight on earlier follow-up, which decreases monotonically to zero at τ, thereby also down-weighting survival differences in late follow-up. The weight for RAH is likewise monotone decreasing, but remains strictly positive at τ, and therefore places relatively greater emphasis on late-stage survival differences than the weight function for D-RMST. This common representation allows us to formally examine the test efficiency in the next section.

2.4 Asymptotic efficiency

To compare the three tests on a single scale we use the *Pitman efficacy*, the standard local-power measure for asymptotically normal tests [Pitman, 1949]. Following Tian et al. [2018], we embed the two-sample problem in a sequence of contiguous (local) alternatives indexed by a fixed log-hazard-ratio shape $\alpha(\cdot)$ on $(0, \tau]$,

$$H_{An}: \qquad \log\left\{\frac{\lambda_{1n}(t)}{\lambda_0(t)}\right\} = \frac{\alpha(t)}{\sqrt{n}}, \tag{9}$$

so that the two hazards coincide as $n \to \infty$, while the between-group difference is in the order of $O(1)$. Under (9), for test $X \in \{\text{Cox}, D\text{-RMST}, \text{RAH}\}$, the test statistic is asymptotically normal, i.e.,

$$\sqrt{n}\, U_X \xrightarrow{d} N\big(\mu_X,\ \sigma_X^2\big),$$

where

$$\mu_X = \int_0^\tau W_X(t)\, \alpha(t)\, \lambda_0(t)\, dt \qquad \text{and} \qquad \sigma_X^2 = \int_0^\tau \frac{W_X(t)^2}{w(t)}\, \lambda_0(t)\, dt.$$

The drift μ_X is obtained by substituting $\sqrt{n}\hat{\Delta}_\Lambda(t)$ by $\alpha(t)\lambda_0(t)$ in (7). The variance $\hat{\sigma}_X^2$ is the null variance of (7). The *asymptotic efficiency* (AE), or Pitman efficacy, of test X is the squared standardized drift

$$\xi_X = \frac{\mu_X^2}{\sigma_X^2}, \tag{10}$$

which governs the local power. The derivation of μ_X for RAH-based test can be found in Appendix A. Furthermore, the explicit expressions of ξ_{Cox}, $\xi_{D\text{-RMST}}$ and ξ_{RAH} are provided in Appendix A.2. The *asymptotic relative efficiency* (ARE) of test X versus test Y is the ratio of corresponding AEs:

$$\text{ARE}(X, Y) = \frac{\xi_X}{\xi_Y}, \tag{11}$$

which approximates the inverse ratio of the sample sizes the two tests require to reach the same local power against (9); $\text{ARE}(X, Y) > 1$ means that test X is more efficient and needs proportionally fewer subjects than Y [Tian et al., 2018, Pitman, 1949]. Because ARE

may be a poor proxy for the sample size ratio in finite-sample setting, we also report the *empirical relative efficiency* (ERE), the ratio of squared standardized effect sizes estimated by simulation (Section 4),

$$\mathrm{ERE}(X, Y) = \frac{E(U_X)^2/\mathrm{Var}(U_X)}{E(U_Y)^2/\mathrm{Var}(U_Y)}, \tag{12}$$

with the expectations and variances approximated by their Monte Carlo averages in simulation [Tian et al., 2018].

3 The geometric interpretation of AE

Section 2.3 reduced all three tests to the weighted logrank test (7) and showed that their AE depends on the test only through the weight function. This is the basis of the classical comparison of Tian et al. [2018], who contrast the Cox and D-RMST tests by inspecting their weight functions $W_{\mathrm{Cox}}(\cdot)$ and $W_{D\text{-RMST}}(\cdot)$ directly. In this section, we replace the side-by-side weight comparison with a geometric quantity, a cosine, turning AE into an angle in a function space.

3.1 Signal direction and test direction

Suppose that all candidate statistics are evaluated on the same horizon τ, and consider the function space $L^2([0, \tau], \lambda_0(v)\, dv)$, where the inner product

$$\langle f, g \rangle = \int_0^\tau f(v)\, g(v)\, \lambda_0(v)\, dv, \tag{13}$$

and norm $\|f\|^2 = \langle f, f \rangle$. Next, we define the *signal direction* for local alternative $\alpha(\cdot)/\sqrt{n}$ and *test direction* for test $X \in \{\mathrm{Cox}, D\text{-RMST}, \mathrm{RAH}\}$, as

$$s_\alpha(t) = \alpha(t)\sqrt{w(t)} \qquad \text{and} \qquad h_X(t) = \frac{W_X(t)}{\sqrt{w(t)}}, \tag{14}$$

respectively.

The signal direction, $s_\alpha(t)$, describes where the detectable treatment effect is by combining the local log-HR representing the underlying between-group difference at t and $\sqrt{w(t)}$ representing the amount of information available to detect the difference at time t. Thus, $s_\alpha(t)$ is large where the true effect is big and sufficient information is available to detect it. It is a "direction" since only its shape over t, not its overall magnitude, matters. The test direction, $h_X(t) = W_X(t)/\sqrt{w(t)}$, is the analogous object built from the test's own weight. $W_X(\cdot)$ records where the test is prepared to look for the effect. $\sqrt{w(t)}$ in the denominator ensures that $h_X(t)$ represents test weight per unit of information available at time t. Next, we have

$$\langle s_\alpha, h_X \rangle = \int_0^\tau \alpha(t)\sqrt{w(t)}\, \frac{W_X(t)}{\sqrt{w(t)}}\, \lambda_0(t)\, dt = \int_0^\tau W_X(t)\, \alpha(t)\, \lambda_0(t)\, dt = \mu_X,$$

$$\|h_X\|^2 = \int_0^\tau \frac{W_X(v)^2}{w(v)}\,\lambda_0(v)\,dv = \sigma_X^2,$$

Consequently the AE (10)

$$\xi_X = \frac{\langle s_\alpha, h_X\rangle^2}{\|h_X\|^2}. \tag{15}$$

3.2 The cosine metric

Writing the inner product in (15) as

$$\langle s_\alpha, h_X\rangle = \|s_\alpha\|\,\|h_X\|\cos(\theta_X),$$

or equivalently,

$$\cos^2(\theta_X) = \frac{\langle s_\alpha, h_X\rangle^2}{\|s_\alpha\|^2\|h_X\|^2} = \frac{\xi_X}{\|s_\alpha\|^2}, \tag{16}$$

where θ_X denotes the angle between the signal direction s_α and the test direction h_X. Intuitively, θ_X measures the mismatch between where the test places its weight and where the detectable treatment effect is concentrated: a small θ_X indicates good alignment and high local power. The quantity $\|s_\alpha\|^2$ is the AE of the oracle test using the optimal weight $\alpha(t)w(t)$. Therefore, $\cos^2(\theta_X)$ is the ARE of test X relative to the oracle test, O. In general, the ARE of test X vs test Y can be expressed as

$$\mathrm{ARE}(X, Y) = \frac{\cos^2(\theta_X)}{\cos^2(\theta_Y)} = \frac{\mathrm{ARE}(X, O)}{\mathrm{ARE}(Y, O)}. \tag{17}$$

As a preview, Figure 1 provides geometric representations for three data-generating processes considered in [Tian et al., 2018]. Specifically, the left column displays $s_\alpha(t)$, $h_{\mathrm{Cox}}(t)$, $h_{\mathrm{DRMST}}(t)$, and $h_{\mathrm{RAH}}(t)$, whereas the right column depicts the corresponding directions on a compass, with the signal direction fixed at 12 o'clock and each test's $\cos^2\theta_X$ shown at the tip of the corresponding arrow for test direction. More detailed results are presented in Section 4.

The cosine metric naturally applies at the trial design stage. Suppose that one has access to:

- assumed survival functions $S_0(t)$ and $S_1(t)$;
- assumed treatment-effect $\alpha(t)$ (e.g., PH, delayed benefit, et al.);
- planned censoring and accrual distribution (which affects $w(\cdot)$); and
- a set of test statistics.

Given these inputs, one can compute the signal and test directions and $\cos^2(\theta_X)$ for each candidate test and select the test with the highest $\cos^2(\theta_X)$, the one most aligned with the expected signal. Crucially, censoring and accrual are part of trial design and affect the value of $\cos^2(\theta_X)$. By varying $S_C(t)$, one can trace how the cosine metric changes and understand whether the selected test is robust to design assumptions. We demonstrate this in Section 5 for CheckMate 057, and specifically trace the preferred test across design choices in a sensitivity analysis.

4 Simulation Study

4.1 Design

We compare the Cox, D-RMST, and RAH-based tests via ARE and ERE using the same data-generating process of Tian et al. [2018]. The baseline event-time distribution is Weibull(κ, η_0) with scale parameter η_0 chosen so that $S_0(40)$ equals a specified target survival rate at $\tau =$ 40 months and the default shape parameter $\kappa = 0.826$ yielding a decreasing hazard $\lambda_0(t) \propto t^{-0.174}$. Censoring time $C = \min(E, U)$, where $E \sim \text{Exp}(\lambda_c)$ and $U \sim \text{Uniform}(24, 43)$ represent loss of follow-up and random entry, respectively. Two rates $\lambda_c \in \{0.0006, 0.025\}$ for E yield light ($S_C(40) = 0.154$) and heavy ($S_C(40) = 0.058$) censoring. Sample size $n_0 = n_1 = 400$ per arm; $B = 2000$ replicates are used for each simulation. We consider the following alternatives:

PH scenario (Case 1): The event-time distribution in the treatment group is Weibull(κ, η_1), where $\eta_1 = \eta_0/0.70^{1/\kappa}$ yields a HR of 0.70. η_0 is adjusted so that $S_0(40) = 0.10, \ldots, 0.80$ and 0.9.

Non-PH scenarios: $\kappa = 0.826$ and $S_0(40) = 0.50$ for control subjects. The log-HR $\alpha(t) = \log[\lambda_1(t)/\lambda_0(t)]$ follows two patterns:

- **Case 2 (fading benefit):** $\alpha(t) = -c_0 \max(43 - t,\, 0)^s, s \in \{0.5, 1, 2\}$. The treatment benefit is the biggest early and fades to null at month 43.
- **Case 3 (delayed benefit):** $\alpha(t) = -c_0 t^s, s \in \{0.5, 1, 2\}$. The treatment benefit is absent at $t = 0$ and increases over follow-up.

In both cases $c_0 > 0$ is selected so that the mean log-HR over $[0, \tau]$ is approximately $\log(0.70)$. For Cases 2 and 3, event-times in treatment arm are generated by numerical inversion of the cumulative hazard, i.e., $T_1 \sim \Lambda_1^{-1}\{\text{Exp}(1)\}$.

Figures 2, 3, and 4 display the survival curves and HRs for three cases. Each figure uses a four-panel layout: the top-left panel shows $\exp\{\alpha(t)\}$ (the HR), and the remaining panels show $S_0(t)$ (control, solid) against $S_1(t)$ (treatment, dashed) for different values of s.

Three simulation settings are also studied:

1. the PH setting with a HR=0.82 (Appendix C);
2. the PH setting with alternative baseline survival distributions (Weibull(κ, η_0) with $\kappa = 0.7, 1$ and 1.5), sample size (150 per arm), and trial horizon ($\tau = 10$) month (Appendix D);
3. the crossing-hazard setting with $\alpha(t) = \log(0.46) + (t/43)$ (the HR begins below 1 and rises to 1.25 at month 43) (Appendix E).

4.2 PH scenario (Case 1)

Table 1 summarizes the comparison for $S_0(40) \in \{0.10, \ldots, 0.90\}$, $\kappa = 0.826$, and HR $= 0.70$. Under light censoring, the ARE of the D-RMST-based test relative to Cox decreases gradually as $S_0(40)$ increases. Under heavy censoring, however, this ARE remains close to 1 across the full range of $S_0(40)$.

The ARE of the RAH-based test relative to the Cox model-based test is also below 1 throughout. This is expected as the Cox model-based test is optimal under the PH assumption. The efficiency declines as $S_0(40)$ increases and more strongly under heavy censoring. As $S_0(\tau)$ increases, the lower bound $W_{\mathrm{RAH}}(\tau) = S_0(\tau)$ becomes larger, so that $h_{\mathrm{RAH}}(\tau) = 2\sqrt{S_0(\tau)/S_C(\tau)}$ also increases. This inflates $\|h_{\mathrm{RAH}}\|^2$ and enlarges RAH's discrepancy from the optimal test direction. At $S_0(40) = 0.10$ under heavy censoring, ARE(RAH/Cox) is 0.798, whereas ARE(RAH/D-RMST) is 0.812. The ordering

$$\text{Cox} \geq \text{RAH} \approx D\text{-RMST}$$

under PH with light censoring and small $S_0(\tau)$ is also evident geometrically in Figure 1 (top row): the signal direction is aligned with the Cox model-based test direction, whereas the RAH and D-RMST test directions are more off-axis.

The ARE of the RAH-based test relative to the D-RMST-based test is close to 1 under light censoring but substantially lower under heavy censoring, indicating a more pronounced performance separation between the two procedures.

Table 2 reports results for $S_0(40) = 0.50$ and $\kappa \in \{0.5, 0.826, 1, 1.5, 2\}$. Under light censoring, the ARE of the RAH-based test relative to the Cox model-based test decreases from 0.922 when $\kappa = 0.5$, corresponding to a decreasing baseline hazard, to 0.737 when $\kappa = 2$, corresponding to an increasing baseline hazard. Intuitively, an increasing hazard shifts more events toward later follow-up, where RAH weighting differs more substantially from optimal weighting of Cox model-based test, thereby reducing its efficiency.

Both Tables 1 and 2 show systematically larger efficiency losses for the RAH-based test under heavy censoring than under light censoring. This pattern persists across alternative baseline survival distributions and trial horizons.

In summary, these results show that the ARE of the competing tests depends materially on the censoring distribution. The censoring and accrual patterns induced by the study design should not be viewed merely as nuisance features; rather, they are important design inputs for choosing testing procedures.

4.3 Non-PH: fading benefit (Case 2)

Table 3 presents the two monotone non-PH scenarios with $\kappa = 0.826$, $S_0(40) = 0.50$, and $s = 1$. Under Case 2, the D-RMST-based test outperforms the RAH-based test, with the Cox model-based test performing intermediately, i.e.,

$$D\text{-RMST} \geq \text{Cox} \geq \text{RAH}.$$

Because the treatment effect is concentrated in the early follow-up period, the down-weighting of late differences by D-RMST causes little efficiency loss. In contrast, the RAH-based test

Table 1: ARE and ERE under PH (HR= 0.70), $\kappa = 0.826$, $\tau = 40$, $n = 400$ per arm, $B = 2000$.

Censoring	$S_0(40)$	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	0.10	0.994	0.941	0.947	1.006	0.946	0.940
Light	0.30	0.967	0.898	0.929	0.978	0.860	0.879
Light	0.50	0.942	0.871	0.924	0.948	0.893	0.942
Light	0.70	0.920	0.851	0.924	0.912	0.866	0.950
Light	0.90	0.901	0.835	0.927	0.919	0.832	0.905
Heavy	0.10	0.982	0.798	0.812	0.980	0.725	0.739
Heavy	0.30	0.997	0.720	0.722	1.019	0.691	0.678
Heavy	0.50	0.998	0.680	0.682	1.024	0.671	0.656
Heavy	0.70	0.995	0.654	0.658	1.011	0.677	0.670
Heavy	0.90	0.990	0.635	0.642	1.019	0.658	0.646

Table 2: ARE and ERE under PH (HR= 0.70), $S_0(40) = 0.50$, $\tau = 40$, $n = 400$ per arm, $B = 2000$.

Censoring	κ	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	0.500	0.962	0.922	0.959	0.970	0.914	0.942
Light	0.826	0.942	0.871	0.924	0.948	0.893	0.942
Light	1.000	0.934	0.844	0.904	0.953	0.838	0.879
Light	1.500	0.919	0.781	0.850	0.936	0.779	0.832
Light	2.000	0.914	0.737	0.806	0.917	0.738	0.805
Heavy	0.500	0.996	0.750	0.753	1.016	0.737	0.725
Heavy	0.826	0.998	0.680	0.682	1.024	0.671	0.656
Heavy	1.000	0.998	0.653	0.654	1.025	0.643	0.628
Heavy	1.500	0.994	0.599	0.603	1.015	0.577	0.569
Heavy	2.000	0.991	0.570	0.576	1.029	0.536	0.521

continues to assign positive weight to late follow-up times, where the treatment effect has largely disappeared. This leads to the expected ordering, D-RMST $\geq$ Cox $\geq$ RAH. The corresponding geometric interpretation is shown in the middle row of Figure 1.

4.4 Non-PH: delayed benefit (Case 3)

Under Case 3, the ARE of the RAH-based test relative to the Cox model-based test exceeds 1 under both censoring regimes. The treatment effect is concentrated at late time, whereas the weight for D-RMST approaches zero near τ, effectively down-weighting the events in the most informative time window. The Cox model also assigns decreasing weight to later follow-up times, as its weight function $w(t)$ declines, but this down-weighting is less pronounced than that of D-RMST. In contrast, the RAH-based test maintains a bigger positive weight near τ, as $S_0(\tau) > S_0(\tau)S_C(\tau) > 0$. Therefore, it preserves a greater alignment with the delayed treatment effect, and yields the expected ordering,

$$\text{RAH} \geq \text{Cox} \geq \text{D-RMST}.$$

See Figure 1 (bottom row) for the corresponding geometric interpretation.

Across the reported simulations, ARE and ERE generally agree in direction; disagreements occur only for comparisons close to 1 (within about 0.07), where Monte Carlo variation and finite-alternative effects can reverse the side of 1.

Table 3: ARE and ERE for monotone non-PH patterns; $\tau = 40$, $S_0(40) = 0.50$, $t_M = 43$, $n = 400$ per arm, $B = 2000$.

Pattern	Cens.	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
fading	Light	1.108	0.741	0.669	1.117	0.707	0.633
fading	Heavy	1.011	0.499	0.494	1.039	0.472	0.454
delayed	Light	0.626	1.190	1.901	0.638	1.179	1.850
delayed	Heavy	0.962	1.349	1.402	0.973	1.375	1.413

5 CheckMate 057: Analysis and Sensitivity

In this section, we illustrate the practical performance of three two-sample tests using data from the CheckMate 057 trial [Borghaei et al., 2015]. To this end, we fit Weibull model to the trial data, compute the theoretical ARE for each test, and conduct simulations to compute the empirical power and required sample sizes under the fitted parametric model.

5.1 Trial background and survival model

CheckMate 057 was a randomized, open-label, Phase III trial comparing nivolumab with docetaxel in 582 patients with previously treated non-squamous non-small-cell lung cancer; the primary endpoint was overall survival [Borghaei et al., 2015]. Because individual-level data are not publicly available, we reconstructed them from the published Kaplan–Meier curves and the reported risk-set information using the algorithm of Guyot et al. [2012], yielding $n_0 = 290$ docetaxel and $n_1 = 292$ nivolumab patients with 419 observed deaths (226 and 193, respectively).

Throughout this section, all three tests use the same analysis horizon $\tau = 24$ months, corresponding to the clinically natural two-year overall-survival landmark and lying within the observed follow-up of both arms. The D-RMST and RAH statistics are defined on $(0, \tau]$ by construction; for the Cox-based test, follow-up is administratively censored at τ before model fitting, so that all three tests read the same data window.

The Schoenfeld global test yields $p = 0.002$, providing strong evidence against the PH assumption. We therefore fit separate Weibull models to each arm and obtain:

$$\begin{aligned} \text{Docetaxel (arm 0):} &\quad \kappa_0 = 1.264,\ \eta_0 = 13.076 \text{ mo}, \\ \text{Nivolumab (arm 1):} &\quad \kappa_1 = 0.987,\ \eta_1 = 17.320 \text{ mo}. \end{aligned}$$

Figure 5 overlays the Kaplan–Meier curves with the fitted Weibull curves, demonstrating their close agreement.

In the docetaxel arm, $\kappa_0 > 1$ induces a monotone increasing hazard; in the nivolumab arm, $\kappa_1 \approx 1$ indicates an approximately constant hazard. The fitted log hazard ratio is

time varying: $\alpha(1) = 0.187$, it crosses zero near month 1.97, and it reaches -0.693 at month 24. This pattern is characteristic of immunotherapy and is consistent with the Schoenfeld diagnosis (Figure 5), whose negative residual slope indicates a decreasing log hazard ratio over time.

For the design-stage evaluations that follow (theoretical ARE, empirical relative efficiency, power, and required sample size), we pair the fitted Weibull event-time model with a censoring working model informed by the trial design. Patients were enrolled between November 2012 and December 2013, and overall survival was assessed at a database lock in March 2015, with a reported minimum follow-up of 13.2 months [Borghaei et al., 2015]. Assuming approximately uniform accrual, we use the rounded administrative follow-up window $U \sim \text{Uniform}(13, 28)$ months. We add an independent exponential dropout time $E \sim \text{Exp}(\lambda_c)$ with $\lambda_c = 0.005$ per month (about 6% per year), and set $C = \min(U, E)$. The administrative window is consistent with the predominantly late censoring in the reconstructed data; the dropout rate is treated as a working design assumption and examined in the sensitivity analysis below.

Table 4 presents the ARE and ERE results. Both measures yield the same ordering of the three tests:

$$\text{RAH} \geq \text{Cox} \geq D\text{-RMST}.$$

Figure 6 presents the compass diagram for the CheckMate 057 data. The corresponding squared cosine similarities are

$$\cos^2(\theta_{\text{Cox}}) = 0.558, \quad \cos^2(\theta_{D\text{-RMST}}) = 0.518, \quad \cos^2(\theta_{\text{RAH}}) = 0.673.$$

$$\cos^2(\theta_{\text{Cox}}) = 0.603, \quad \cos^2(\theta_{\text{D-RMST}}) = 0.469, \quad \cos^2(\theta_{\text{RAH}}) = 0.643.$$

Because the ARE between two tests is given by the ratio of their squared cosine similarities, these values imply

$$\text{ARE}(\text{RAH}/\text{Cox}) = 1.207 \quad \text{and} \quad \text{ARE}(\text{RAH}/D\text{-RMST}) = 1.300.$$

$$\text{ARE}(\text{RAH}/\text{Cox}) = 1.067 \quad \text{and} \quad \text{ARE}(\text{RAH}/\text{D-RMST}) = 1.372.$$

Figure 6 also provides a geometric explanation for why the crossing-hazards alternative favors the RAH-based test. The signal direction places substantial weight on the late follow-up period, where the treatment benefit of nivolumab is greatest. In contrast, the Cox test direction places relatively greater weight on earlier times and gradually decays over follow-up. The D-RMST test direction tapers more rapidly to zero as time approaches τ, thereby placing even less weight on the late follow-up period. The RAH test direction, by comparison, is bounded below by $S_0(\tau) \approx 0.12$, allowing it to retain meaningful weight at later times. Consequently, the RAH test direction is more closely aligned with the signal direction, resulting in greater efficiency under this crossing-hazards alternative.

Beyond the efficiency metrics themselves, a key practical question is how differences in ARE translate into differences in power and required sample size. Figure 7 shows the empirical power as a function of sample size per arm, with the RAH-based test consistently achieving the highest power, followed by the Cox model-based test and then the D-RMST-based test. Table 5 summarizes power at $n_0 = n_1 = 300$ per arm and the first simulated sample size on the prespecified 50-patient grid to reach 80% power. RAH reaches the target

Table 4: Theoretical ARE and empirical ERE under the CheckMate 057 fitted non-PH DGP; all methods use $\tau = 24$ months, $n = 300$ per arm, $B = 500$. Administrative censoring is $U \sim \text{Uniform}(13, 28)$ months, independently combined with exponential dropout.

Comparison	ARE	ERE
D-RMST/Cox	0.928 0.778	0.935 0.776
RAH/Cox	1.207 1.067	1.332 1.167
RAH/D-RMST	1.300 1.372	1.425 1.504

at 300250 patients per arm, compared with 400300 for Cox and 400 for D-RMST. Thus, Cox and D-RMST require 33% and 3320% and 60% more patients than RAH, respectively, on this grid. At 300 patients per arm, empirical power is 8489% for RAH, 7285% for Cox, and 6872% for D-RMST.

Table 5: Empirical power at $n = 300$ per arm and first simulated sample size reaching 80% power (50-patient grid, $B = 2000$), with simulated, ARE-, and ERE-implied ratios relative to RAH. Administrative censoring is $U \sim \text{Uniform}(13, 28)$ months, independently combined with exponential dropout.

Method	Power	n_{80}	Sim. ratio	ARE ratio	ERE ratio
Cox	0.717 0.849	400 300	1.33 1.20	1.21 1.07	1.33 1.17
D-RMST	0.680 0.725	400	1.33 1.60	1.30 1.37	1.43 1.50
RAH	0.842 0.889	300 250	1.00	1.00	1.00

We now assess the robustness of the ordering RAH $\geq$ Cox $\geq$ D-RMST to the analysis horizon τ, dropout rate λ_c, and assumed alternative. The results are reported in Table 6.

1. **Horizon τ:** ARE(D-RMST/Cox) increases substantially with τ, from 0.6640.534 at $\tau = 18$ to 0.9280.778 at $\tau = 24$. The relatively low efficiency of D-RMST at shorter horizons reflects truncation of the late treatment benefit. In contrast, ARE(RAH/Cox) is comparatively stable (1.206, 1.219, and 1.2071.063, 1.080, and 1.067 at 18, 21, and 24 months), while ARE(RAH/D-RMST) decreases from 1.8171.990 to 1.3001.372.

2. **Dropout rate:** Increasing λ_c from 0.001 to 0.020 shifts ARE(D-RMST/Cox) from 0.8970.749 to 1.0530.891, indicating that heavier dropout reduces the relative efficiency of Cox. ARE(RAH/Cox) increases modestly from 1.1871.047 to 1.2711.136, whereas ARE(RAH/D-RMST) decreases from 1.3241.398 to 1.2071.274. This pattern reflects the growing attenuation of the Cox weight at later times as $S_C(t)$ decreases.

3. **PH vs. non-PH:** Under a PH alternative (HR= 0.75, with the same Weibull shape as for docetaxel), Cox is the most efficient, with ARE(D-RMST/Cox)= 1.005 = 0.988 and ARE(RAH/Cox)= 0.875= 0.909. This highlights that RAH's advantage in the fitted non-PH setting arises from its greater alignment with the late treatment benefit; the positive weight floor provides no such advantage under PH.

In summary, the ERE and ARE results are generally consistent. Under the assumed non-PH alternative, the RAH-based test is more efficient than both Cox model-based and

D-RMST-based tests. Under the PH alternative, however, the RAH-based test is slightly less efficient than the Cox model-based test.

Table 6: Sensitivity to analysis horizon, dropout rate, and hazard structure; the common baseline is $\tau = 24$, $\lambda_c = 0.005$, fitted non-PH, $n = 300$ per arm and $B = 500$. Administrative censoring is $U \sim \text{Uniform}(13, 28)$ months, independently combined with exponential dropout.

Panel	Setting	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Horizon	$\tau = 18$	0.664 0.534	1.206 1.063	1.817 1.990	0.607 0.458	1.292 1.102	2.127 2.407
Horizon	$\tau = 21$	0.802 0.657	1.219 1.080	1.519 1.644	0.768 0.613	1.299 1.126	1.691 1.838
Horizon	$\tau = 24$	0.928 0.778	1.207 1.067	1.300 1.372	0.935 0.776	1.332 1.167	1.425 1.504
Dropout	$\lambda_c = 0.001$	0.897 0.749	1.187 1.047	1.324 1.398	0.895 0.723	1.223 1.100	1.367 1.521
Dropout	$\lambda_c = 0.005$	0.928 0.778	1.207 1.067	1.300 1.372	0.935 0.776	1.332 1.167	1.425 1.504
Dropout	$\lambda_c = 0.020$	1.053 0.891	1.271 1.136	1.207 1.274	1.088 0.876	1.369 1.206	1.258 1.377
Hazard	Non-PH (fitted)	0.928 0.778	1.207 1.067	1.300 1.372	0.935 0.776	1.332 1.167	1.425 1.504
Hazard	PH (HR= 0.75)	1.005 0.988	0.875 0.909	0.871 0.919	0.990 0.938	0.904 0.926	0.913 0.988

6 Discussion

The results provide a unified view of the relative efficiency of the Cox-, D-RMST-, and RAH-based tests. A central finding is that no single test is uniformly most efficient; rather, relative efficiency depends jointly on the pattern of treatment effect and the trial design. The Cox HR-based test is most efficient under the PH assumption. Under non-PH alternatives, the D-RMST-based test can be more efficient when treatment effects are stronger earlier in follow-up, whereas the RAH-based test can be more efficient when treatment benefits emerge later. These patterns, however, are not determined by the treatment-effect pattern alone. Censoring and accrual affect the information available over follow-up and can alter, and in some settings reverse, the efficiency ordering among the tests.

The proposed cosine metric provides a geometric way to understand these differences by quantifying the alignment between the treatment-effect signal and the direction targeted by each test, while incorporating the information available over follow-up under the specified trial design. At the design stage, the metric can be used to evaluate candidate tests across a range of plausible assumptions about the treatment-effect pattern and other design features. Such scenario-based comparisons can inform the prespecified choice of the primary test and assess how robust that choice is to uncertainty in the design assumptions.

Several extensions warrant future investigation. The cosine metric could potentially be used not only to compare candidate tests but also as an objective function for optimizing design features such as accrual and follow-up duration. Another extension is incorporate the RAH weight W_{RAH}, which is not a Fleming–Harrington weight [Fleming and Harrington, 1991], into a Max-combo test together with W_{Cox} and $W_{D\text{-RMST}}$. Such a procedure could provide sensitivity to a broader range of treatment-effect patterns, although its operating characteristics would require futher study.

A Asymptotic Distribution of the RAH Statistic

A.1 Local-alternative drift

Throughout we work under *Pitman contiguous alternatives*:

$$H_{An}: \quad \log \frac{\lambda_{1n}(t)}{\lambda_0(t)} = \frac{\alpha(t)}{\sqrt{n}}, \tag{18}$$

where $\alpha(\cdot)$ is a fixed function on $(0, \tau]$. Under (18), to first order in $n^{-1/2}$: $\sqrt{n}[S_{1n}(t) - S_0(t)] \to -S_0(t) \int_0^t \lambda_0(u)\alpha(u)\, du$.

Proposition 1. *Under H_{An}, $\sqrt{n}\log(\hat{\theta}_{\mathrm{RAH}})$ is asymptotically normal with mean*

$$\mu_{\mathrm{RAH}} = \int_0^\tau \left[\frac{1}{F_0(\tau)} - \frac{R_0(t)}{R_0(\tau)}\right] \alpha(t)\lambda_0(t)\, dt. \tag{19}$$

Proof. $\sqrt{n}\log(\hat{\theta}_{\mathrm{RAH}}) = U_n + V_n$, where $U_n = \sqrt{n}[\log \hat{F}_1(\tau) - \log \hat{F}_0(\tau)]$ and $V_n = -\sqrt{n}[\log \hat{R}_1(\tau) - \log \hat{R}_0(\tau)]$. Next, we have

$$\begin{aligned} U_n &= F_0(\tau)^{-1}\sqrt{n}[\hat{F}_1 - \hat{F}_0](\tau) + o_p(1) \\ &= F_0(\tau)^{-1}\sqrt{n}[F_1 - F_0](\tau) - F_0(\tau)^{-1}\sqrt{n}[(\hat{S}_1 - S_1) - (\hat{S}_0 - S_0)](\tau) + o_p(1) \\ &= \int_0^\tau \frac{S_0(\tau)}{F_0(\tau)}\alpha(t)\lambda_0(t)\, dt + \eta_{1n} + o_p(1), \end{aligned}$$

and

$$\begin{aligned} V_n &= -R_0(\tau)^{-1}\int_0^\tau \sqrt{n}[\hat{S}_1(t) - \hat{S}_0(t)]\, dt + o_p(1) \\ &= -R_0(\tau)^{-1}\int_0^\tau \sqrt{n}[S_1(t) - S_0(t)]\, dt - R_0(\tau)^{-1}\int_0^\tau \sqrt{n}\left[(\hat{S}_1 - S_1) - (\hat{S}_0 - S_0)\right](t)\, dt + o_p(1) \\ &= R_0(\tau)^{-1}\int_0^\tau S_0(t)A(t)\, dt + \eta_{2n} + o_p(1), \end{aligned}$$

where $A(t) = \int_0^t \lambda_0(u)\alpha(u)\, du$ and $(\eta_{1n}, \eta_{2n})'$ converging in distribution to a mean zero bivariate Gaussian distribution as $n \to \infty$. Furthermore,

$$R_0(\tau)^{-1}\int_0^\tau S_0(t)A(t)\, dt = A(\tau) - \int_0^\tau \lambda_0(t)\alpha(t)\frac{R_0(t)}{R_0(\tau)}\, dt = \int_0^\tau \left[1 - \frac{R_0(t)}{R_0(\tau)}\right]\alpha(t)\lambda_0(t)dt.$$

Substituting and simplifying gives (19). □

A.2 Pitman efficacy of RAH-based test

The Pitman efficacy formula follows from $\xi_{\mathrm{RAH}} = \mu^2_{\mathrm{RAH}}/\sigma^2_{\mathrm{RAH}}$:

$$\xi_{\mathrm{RAH}} = \frac{\left[\int_0^\tau \left\{\frac{1}{F_0(\tau)} - \frac{R_0(t)}{R_0(\tau)}\right\}\alpha(t)\lambda_0(t)\, dt\right]^2}{\int_0^\tau \left\{\frac{1}{F_0(\tau)} - \frac{R_0(t)}{R_0(\tau)}\right\}^2 \frac{\lambda_0(t)}{w(t)}\, dt}. \tag{20}$$

The efficacies of Cox and D-RMST follow from Tian et al. [2018]:

$$\xi_{\text{Cox}} = \frac{\left[\int_0^\tau w(t)\,\alpha(t)\lambda_0(t)\,dt\right]^2}{\int_0^\tau w(t)\,\lambda_0(t)\,dt}, \tag{21}$$

$$\xi_{D\text{-RMST}} = \frac{\left[\int_0^\tau \{R_0(\tau) - R_0(t)\}\,\alpha(t)\lambda_0(t)\,dt\right]^2}{\int_0^\tau \{R_0(\tau) - R_0(t)\}^2\,\dfrac{\lambda_0(t)}{w(t)}\,dt}. \tag{22}$$

B Derivation of the Weight Function Representations

We prove that each of the three statistics of Section 2.2 is asymptotically equivalent to a version of weighted logrank statistic (7),

$$\int_0^\tau W(t)\,d[\hat{\Lambda}_1(t) - \hat{\Lambda}_0(t)].$$

All derivations use: (i) the KM linear representation

$$\sqrt{n}(\hat{S}_k - S_k) = -S_k\sqrt{n}(\hat{\Lambda}_k - \Lambda_k) + o_p(1);$$

(ii) the functional delta method; and (iii) the local-alternative approximation

$$S_{1n}(t) = S_0(t) - \frac{S_0(t)A(t)}{\sqrt{n}} + o(n^{-1/2}).$$

For simplicity, we also assume equal allocation and common censoring $S_C(\cdot)$.

- **Cox model-based statistic.** Let $\ell_n(\theta)$ be the log partial likelihood over $[0,\tau]$, $U_n(\theta) = \partial\ell_n(\theta)/\partial\theta$ its score, and $\mathcal{I}_n(\theta) = -\partial^2\ell_n(\theta)/\partial\theta^2$ the observed information. The score at $\theta = 0$ is exactly the log-rank statistic, $n^{-1}U_n(0) = U_{\text{Cox}}$ in (8), and

  $$n^{-1}\mathcal{I}_n(0) \longrightarrow \int_0^\tau w(t)\lambda_0(t)dt = \sigma_\theta^{-2}$$

  in probability, uniformly in a neighbourhood of $\theta = 0$. Since $\hat{\theta} = O_p(n^{-1/2})$ under the local alternatives (18) and $U_n(\hat{\theta}) = 0$, a one-term Taylor expansion of $U_n(\theta)$ about $\theta = 0$ gives

  $$0 = n^{-1}U_n(\hat{\theta}) = U_{\text{Cox}} - n^{-1}\mathcal{I}_n(0)\,\hat{\theta} + o_p(n^{-1/2}) \Rightarrow \sqrt{n}\,\hat{\theta} = \sigma_\theta^2\sqrt{n}\,U_{\text{Cox}} + o_p(1).$$

  Hence $\sqrt{n}\,\hat{\theta} = \sigma_\theta^2\sqrt{n}\,U_{\text{Cox}} + o_p(1)$, and Z_{Cox} and the normalized log-rank statistic are asymptotically equivalent. The logrank statistic (8) is already of the weighted-log-rank form, with random weight

  $$\hat{w}_{\text{Cox}}(t) = \frac{Y_1(t)Y_0(t)}{Y_1(t) + Y_0(t)},$$

where $Y_k(t)$ is the size of the risk set of group k at t. By law of large numbers, $n^{-1}\hat{w}_{\text{Cox}}(t)$ converges to $w(t)$, which is $S_0(t)S_C(t)/4$ under equal allocation and common censoring. Therefore, we can set

$$W_{\text{Cox}}(t) = 4w(t) = \ S_0(t)S_C(t),$$

in (7).

- **D-RMST.** By functional delta-method, we have

$$\begin{aligned}
\sqrt{n}\hat{\theta}_{D\text{-RMST}} &= -\sqrt{n}\int_0^\tau S_0(t)\,\hat{\Delta}_\Lambda(t)\,dt + O\big(\sqrt{n}\|\hat{\Delta}_\Lambda\|^2\big) \\
&= -\sqrt{n}\int_0^\tau S_0(t)\int_0^t d\hat{\Delta}_\Lambda(u)\,dt + O_p\big(n^{-1/2}\big) \\
&= -\sqrt{n}\int_0^\tau \left\{\int_u^\tau S_0(t)\,dt\right\} d\hat{\Delta}_\Lambda(u) + O_p\big(n^{-1/2}\big) \\
&= -\sqrt{n}\int_0^\tau \{R_0(\tau) - R_0(u)\}\,d\hat{\Delta}_\Lambda(u) + o_p(1).
\end{aligned}$$

  Therefore, we may set the weight as

$$W_{D\text{-RMST}}(t) = \frac{R_0(\tau) - R_0(t)}{R_0(\tau)},$$

  as $Z_{D\text{-RMST}}$ is invariant to multiplying any nonzero constant to the weight function.

- **RAH.** Using the same argument together with the delta method for $\log F_k(\tau)$ and $\log R_k(\tau)$,

$$\log\hat{\theta}_{\text{RAH}} = \int_0^\tau \left[\frac{S_0(\tau)}{F_0(\tau)} + \frac{R_0(\tau) - R_0(t)}{R_0(\tau)}\right] d\hat{\Delta}_\Lambda(t) + O_p(n^{-1}).$$

  Multiplying by $F_0(\tau) > 0$ gives the equivalent weight

$$W_{\text{RAH}}(t) = S_0(\tau) + F_0(\tau)\,\frac{R_0(\tau) - R_0(t)}{R_0(\tau)} = S_0(\tau) + \big[1 - S_0(\tau)\big]\,W_{D\text{-RMST}}(t).$$

C Simulation Study for PH Setting at HR = 0.82

We repeat the simulation for PH scenarios with HR=0.82. The results are reported in Tables 7 and 8. Both ARE and ERE are similar to those for HR= 0.70:

D Simulation Study for PH Setting with Different Sample Size, Horizon and Survival Distributions

We also repeat simulation study the PH setting under different design: $\tau = 10$, $n_0 = n_1 = 150$, and three baseline distributions (an exponential with $S_0(10) \approx 0.37$; a Weibull

Table 7: ARE and ERE under PH (HR= 0.82), $\kappa = 0.826$, $\tau = 40$, $n = 400$ per arm, $B = 2000$.

Censoring	$S_0(40)$	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	0.10	0.994	0.941	0.947	1.001	0.938	0.937
Light	0.20	0.980	0.917	0.935	0.987	0.898	0.910
Light	0.30	0.967	0.898	0.929	0.968	0.911	0.942
Light	0.40	0.954	0.883	0.926	0.957	0.872	0.911
Light	0.50	0.942	0.871	0.924	0.929	0.885	0.953
Light	0.60	0.931	0.860	0.924	0.958	0.852	0.889
Light	0.70	0.920	0.851	0.924	0.930	0.839	0.903
Light	0.80	0.911	0.842	0.925	0.952	0.831	0.872
Light	0.90	0.901	0.835	0.927	0.913	0.834	0.914
Heavy	0.10	0.982	0.798	0.812	0.990	0.731	0.738
Heavy	0.20	0.993	0.751	0.756	1.004	0.725	0.722
Heavy	0.30	0.997	0.720	0.722	1.004	0.704	0.701
Heavy	0.40	0.998	0.698	0.699	1.004	0.689	0.686
Heavy	0.50	0.998	0.680	0.682	1.007	0.672	0.667
Heavy	0.60	0.997	0.666	0.668	1.001	0.676	0.675
Heavy	0.70	0.995	0.654	0.658	0.996	0.685	0.688
Heavy	0.80	0.992	0.644	0.649	0.993	0.661	0.665
Heavy	0.90	0.990	0.635	0.642	1.054	0.698	0.662

Table 8: ARE, power, and ERE under PH (HR= 0.82), $S_0(40) = 0.50$, $\tau = 40$, $n = 400$ per arm, $B = 2000$.

Censoring	κ	ARE(D/C)	ARE(R/C)	ARE(R/D)	Power C	Power D	Power R	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	0.500	0.962	0.922	0.959	0.459	0.452	0.433	0.980	0.918	0.937
Light	0.826	0.942	0.871	0.924	0.438	0.418	0.395	0.929	0.885	0.953
Light	1.000	0.934	0.844	0.904	0.450	0.406	0.393	0.941	0.838	0.891
Light	1.500	0.919	0.781	0.850	0.416	0.382	0.341	0.931	0.802	0.862
Light	2.000	0.914	0.737	0.806	0.385	0.349	0.307	0.906	0.765	0.844
Heavy	0.500	0.996	0.750	0.753	0.377	0.379	0.307	1.003	0.744	0.741
Heavy	0.826	0.998	0.680	0.682	0.338	0.337	0.260	1.007	0.672	0.667
Heavy	1.000	0.998	0.653	0.654	0.299	0.301	0.237	0.996	0.632	0.635
Heavy	1.500	0.994	0.599	0.603	0.280	0.284	0.224	0.991	0.632	0.638
Heavy	2.000	0.991	0.570	0.576	0.251	0.253	0.183	0.995	0.584	0.587

baseline with a decreasing hazard ($\kappa = 0.7$) and $S_0(10) = 0.5$; and a Weibull baseline with an increasing hazard ($\kappa = 1.5$) and $S_0(10) = 0.5$.) We also consider light censoring with $C \sim \min\{U(8, 12), \text{Exp}(0.02)\}$ and heavy censoring with $C \sim \min\{U(8, 12), \text{Exp}(0.20)\}$.

The results are reported in Table 9. RAH is more efficient than D-RMST under light censoring, whereas D-RMST is substantially more efficient than RAH under heavy censoring.

E Additional Simulations: Crossing Hazards

In this simulation, $\alpha(t) = \log(0.46) + (t/t_M)$ with $t_M = 43$ months, $\kappa = 0.826$, and $S_0(40) = 0.50$: the HR starts at 0.46, crosses one, and rises above 1 late in follow-up. Thus, an early treatment benefit is reversed to harm at late times. Table 10 reports the ARE and ERE under light and heavy censoring. D-RMST is more efficient than Cox, whereas RAH

Table 9: ARE and ERE under PH (HR= 0.70), $\tau = 10$, $n = 150$ per arm, $B = 500$.

Censoring	Baseline	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	Weibull(kappa=0.7)	0.920	0.981	1.067	0.985	0.974	0.989
Light	Exponential(lambda0=0.10)	0.913	0.976	1.068	0.941	0.989	1.051
Light	Weibull(kappa=1.5)	0.853	0.954	1.119	0.895	0.906	1.013
Heavy	Weibull(kappa=0.7)	0.970	0.669	0.690	1.018	0.722	0.709
Heavy	Exponential(lambda0=0.10)	0.973	0.665	0.684	1.014	0.670	0.660
Heavy	Weibull(kappa=1.5)	0.987	0.651	0.659	0.987	0.611	0.619

is less efficient than both Cox and D-RMST, under both censoring regimes. The efficiency separation is larger under heavy censoring.

Table 10: ARE and ERE under crossing hazards; $t_M = 43$, $\tau = 40$, $S_0(40) = 0.50$, $n = 400$ per arm, $B = 2000$.

Cens.	ARE(D/C)	ARE(R/C)	ARE(R/D)	ERE(D/C)	ERE(R/C)	ERE(R/D)
Light	1.193	0.683	0.573	1.273	0.597	0.469
Heavy	1.016	0.429	0.422	1.060	0.374	0.353

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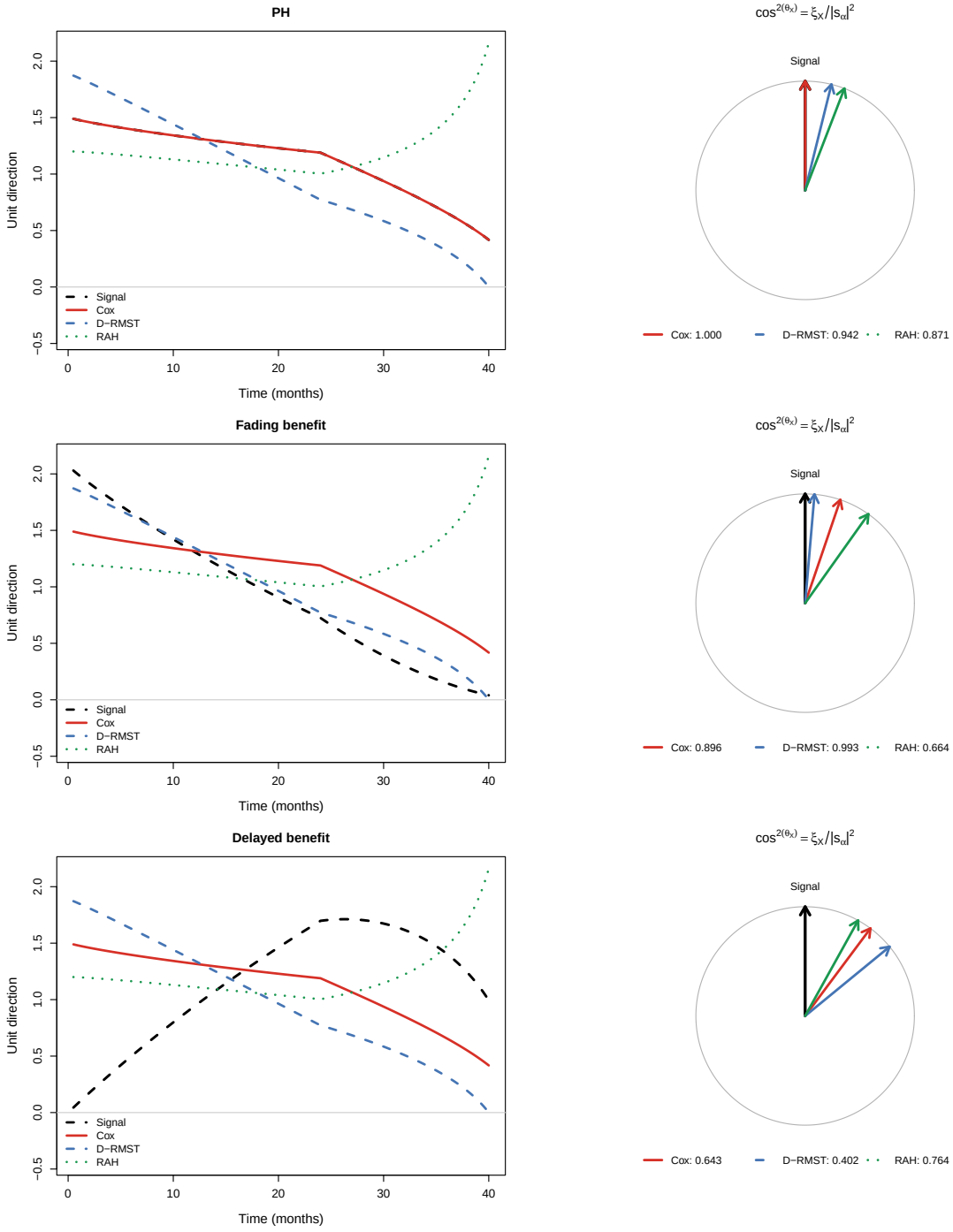


Figure 1: Unit-vector functions (left column) and compass diagrams (right column) for three scenarios (Weibull baseline: $\kappa = 0.826$, $S_0(40) = 0.50$, light censoring). The D-RMST and RAH tests are evaluated at $\tau = 40$ months; Signal direction is placed at 12 o'clock in the compass diagrams; each test arrow is annotated with its $\cos^2(\theta_X)$ value. Under PH (top), Cox is aligned, while D-RMST and RAH lose some efficiencies. Under delayed benefit (bottom), the signal rotates toward the RAH direction, raising $\cos^2(\theta_{\text{RAH}})$ above $\cos^2(\theta_{\text{Cox}})$.

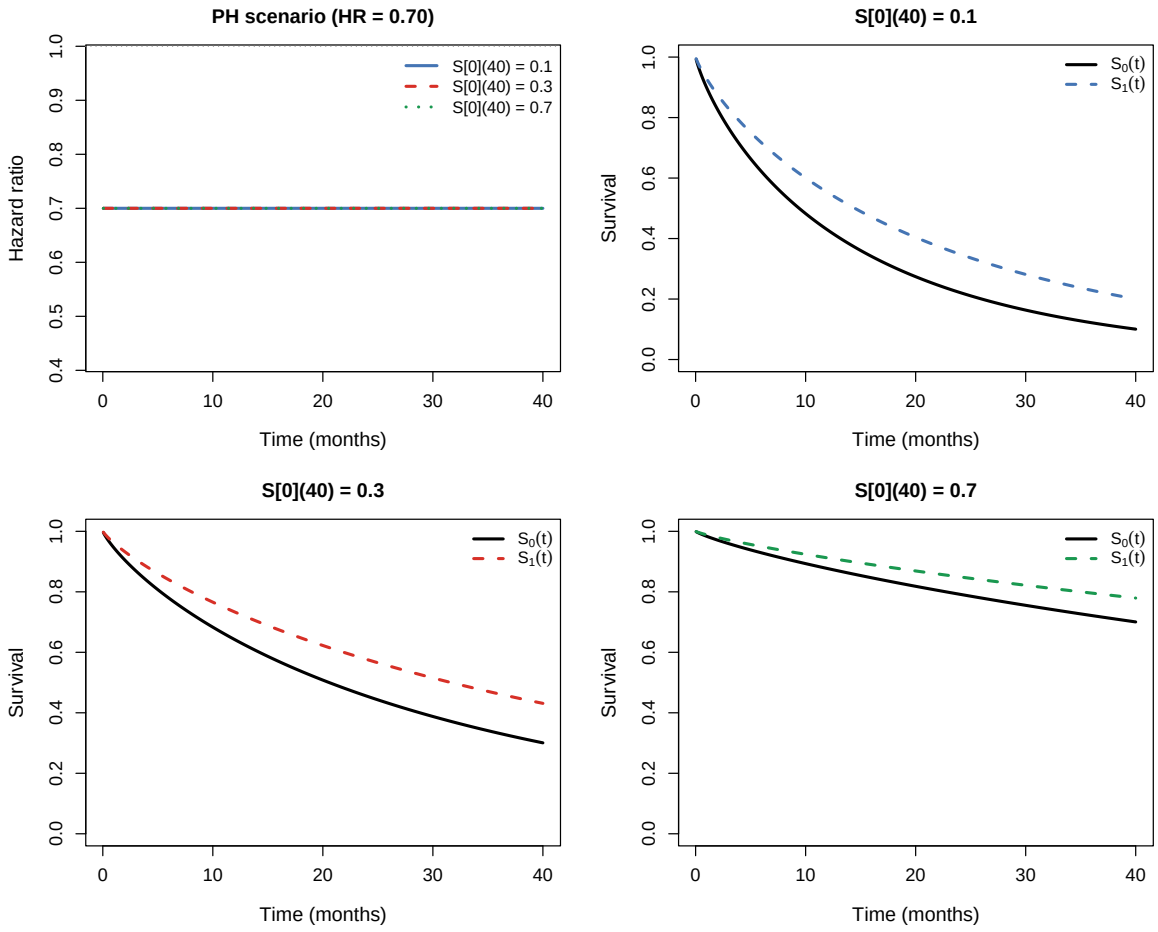


Figure 2: **PH scenario** (HR $= 0.70$) and $S_0(\tau) \in \{0.10, 0.30, 0.70\}$. The HR is flat at 0.70; higher event rates produce greater absolute separation between two survival curves.

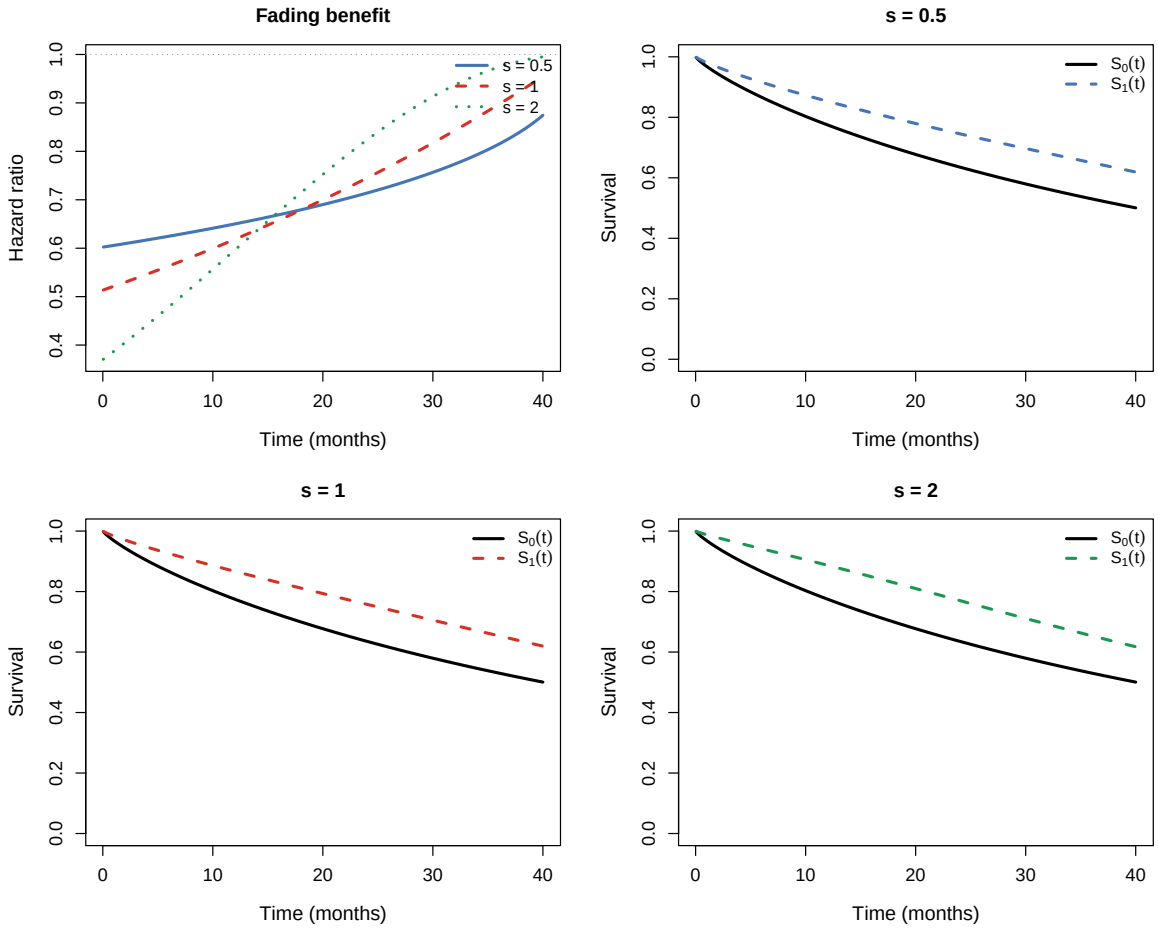


Figure 3: **Case 2: fading benefit.** $\alpha(u) = -c_0 \max(t_M - u, 0)^s$, mean log-HR $= \log(0.70)$. The HR is below 1 early and converges to 1 by t_M; larger s concentrates the benefit closer to $u = 0$.

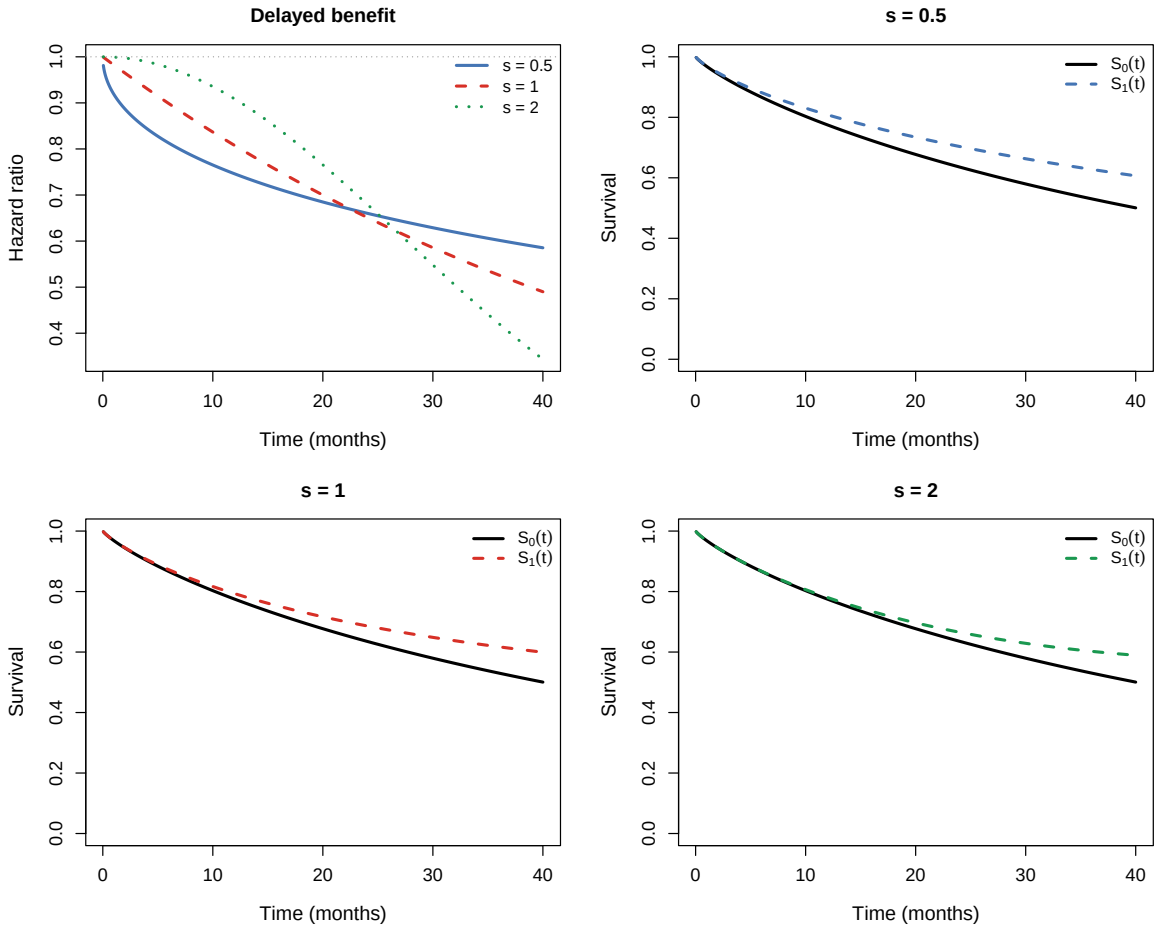


Figure 4: **Case 3: delayed benefit.** $\alpha(u) = -c_0\, u^s$, mean log-HR $= \log(0.70)$. The HR starts at 1 and decreases; larger s delays the onset of the treatment effect, with the benefit concentrated at later time points for $s = 2$.

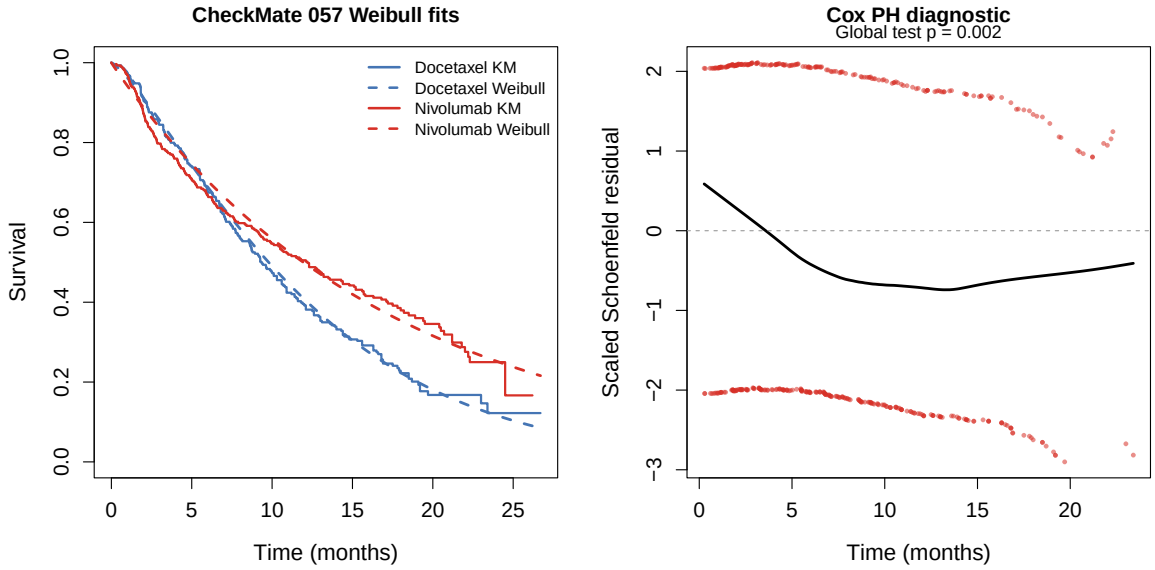


Figure 5: Survival model fit (left) and proportional-hazards diagnostic (right) for the CheckMate 057 trial.

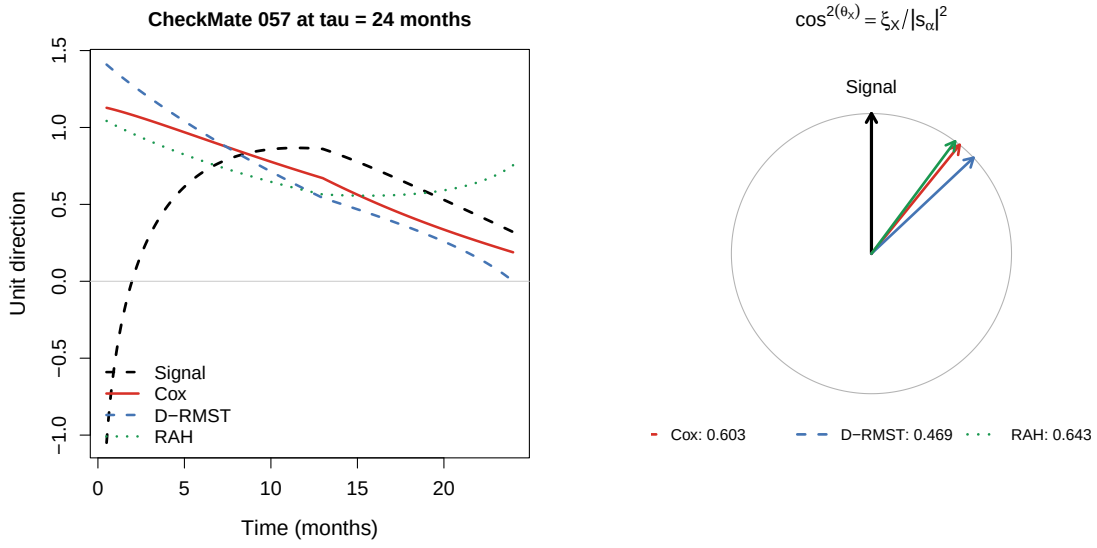


Figure 6: Unit-vector functions (left) and compass diagram (right) for Weibull distributions fitted to CheckMate 057 trial. All three tests use $\tau = 24$ months; the function panel displays them on $[0, \tau]$. The signal direction is placed at 12 o'clock and reflects the crossing-hazards pattern. Each test direction is annotated with $\cos^2(\theta_X)$. RAH is closest to the signal direction, followed by Cox and D-RMST.

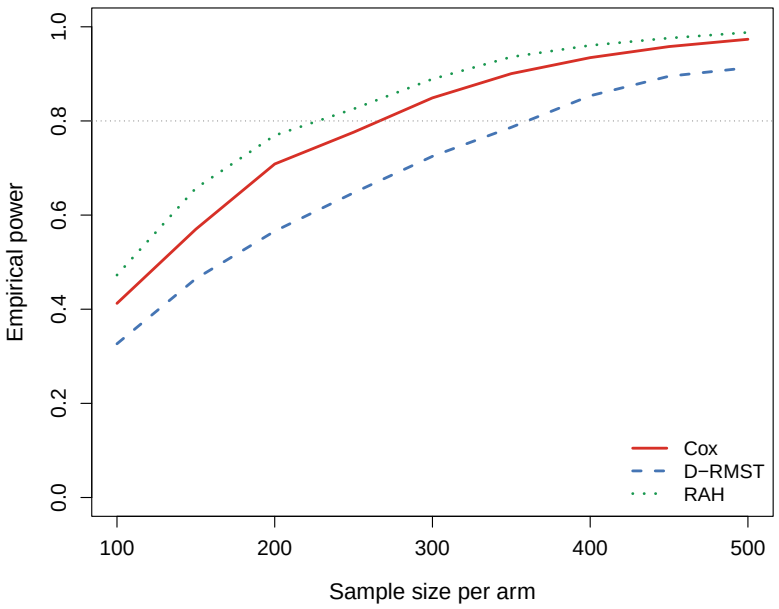


Figure 7: Empirical power for Cox (red, solid), D-RMST (blue, dashed), and RAH (green, dotted) under the fitted CheckMate 057 non-PH alternative. Horizontal line: 80% power.